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Devonian–Carboniferous paleopoles for Africa: consequences for Hercynian geodynamics

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ABSTRACT

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Paleomagnetic analysis of marine formations in the northwestern Sahara (Algeria) yields the following results: three sites in the lower Devonian *Orthoceras*-bearing limestones of Béni-Abbès (29.67° N, 2.15° W) indicate a mean direction of remanence: $D_m = 149.6°$; $I_m = 1.6°$; $\alpha_{95} = 6.8°$. This typical Permian direction implies a complete remagnetization of the sites. The Famennian griotte limestones of the Ben-Zireg anticline (31.92° N, 1.82° W) show high remanent magnetization intensities. In this structure, which has undergone polyphase tectonism, multicomponent analysis evidences a high unblocking temperature (HTb) component carried by haematite leading to a positive fold test (after partial unfolding) at the 95% confidence level with a mean direction $D = 156.4°$, $I = 47.6°$, $\alpha_{95} = 0.9°$ and attributed to the Early Carboniferous (355 Ma) with a paleopole at 25.3° S, 21.1° E, $dp = 0.8°$, $dm = 1.2°$. An intermediate unblocking temperature (ITb) component of magnetization carried probably by magnetite, also present, satisfies a positive fold test (after tilt correction), with a mean direction $D = 154.9°$, $I = 54.3°$, $\alpha_{95} = 3.7°$. This pre-folding direction of magnetization is attributed to the Famennian. It gives a paleopole at 19.2° S, 19.8° E, $dp = 3.7°$, $dm = 5.3°$.

Comparison with previous results shows a significant S–N crustal shortening of about 2500 km during Carboniferous times accompanied by a clockwise rotation of the Saharan region of about 25°. These new results give information about the tectonic phenomena involved in the collision between Gondwanaland and Laurasia: they argue for a Pangea A_2 dominating model, and define mean values for shortening rate and plate velocity comparable to the present ones (3 cm/yr).

Introduction

The northwestern Sahara has yielded several paleopoles for different periods of the Paleozoic (Morel et al., 1981; Daly and Irving, 1983). An Early to Middle Devonian paleopole has been reported for rocks from Mauritania (Kent et al., 1984), but it is possible that these rocks were remagnetized in the Permian, given the location of the paleopole and the absence of any field tests to constrain the age of magnetization. Consequently Devonian paleomagnetic directions from this region, and from Africa in general, are nearly absent. Other paleopoles from Cambro-Ordovician to Middle Devonian have been recently evidenced in Africa (Bachtadse et al., 1987; Hargraves et al., 1987; Saradeth et al., 1989) and in other continents (Schmidt et al., 1987; Li et al., 1988).

Different Apparent Polar Wander Paths (APWP) for Gondwanan have been established through the "Pangea" reconstruction of Smith and Hallam (1970). But two main paths from Morel and Irving (1978) remain viable if we take into account African data (path *X*) or other Gondwanan data (path *Y*).

We have sampled Middle Paleozoic marine formations in a period covering the duration of the

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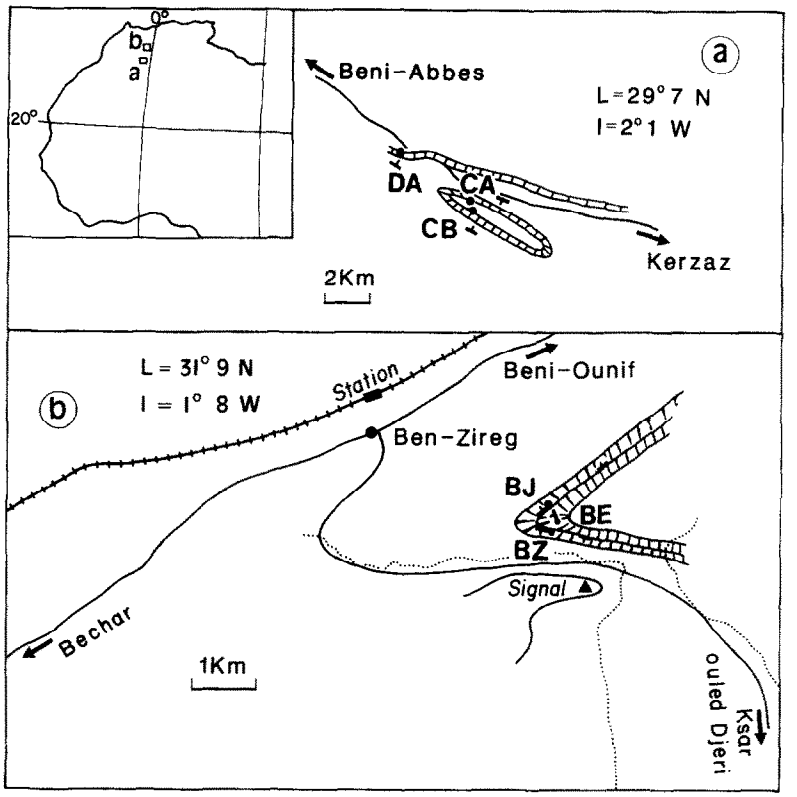


Fig. 1. Location map and sampling sites in northwestern Sahara representing Béni-Abbès monoclinal and synclinal structures (a) and the Ben-Zireg anticline (b).

Hercynian collision along the transect from Béni-Abbès to the north of Béchar (Fig. 1) in order to contribute to the understanding of the continental collision between Gondwanaland and Laurasia which led to the formation of the supercontinent 'Pangea' at the end of the Paleozoic. In this context our paleomagnetic study attempted to improve the understanding of three main problems: the first is linked to the closure of the pre-Hercynian ocean. The idea of a large oceanic domain at the end of the Devonian (Hailwood, 1974) is debated in Salmon et al. (1986), but the exact time of closure is still unknown and constitutes a goal for paleomagnetic measurements on rocks from the Sahara, and subsequent comparisons with data from other cratons. The second subject corresponds to the continental fit between Gondwanaland and Laurasia. The third purpose is to test the hypothesis of a polyphase collisional tectonic regime ranging from Tunisia to Mauritania following a clockwise rotation of the Gondwana block (Matte, 1986).

Geological setting

Since very Early Paleozoic times, the Saharan platform has been covered by sandstones and/or shales from terrigeneous sources. During the Devonian, sedimentary facies changed completely. Large platform carbonate deposits argue for a position of the northern Sahara at low latitudes. The orogenic trend of this region defines an important curvature: the Anti-Atlas–Ougarta Arc (Donzeau, 1974). This structure follows the Reguibat cratonic border, with a western branch corresponding to the Anti-Atlas mean direction (N50°), and an eastern branch corresponding to the Ougarta, mean orientation (N120°). The tectonic evolution was quite different between the northern and the southern part of the area studied.

A tectonic phase located at the Devonian–Carboniferous boundary (360 Ma, following Haq and Van Eysinga, 1987) is well documented and accurately dated by biostratigraphy, especially in the Ben-Zireg structure (Pareyn, 1961), where the absence of latest Famennian strata and a Lower Carboniferous (Tournaisian) unconformity have been recorded. Tectonic activity continued during the Visean with another uncomformity and a gap in sedimentation. In the southern part (Béni-Abbès region), the tectonic phases occur later with a belated Hercynian phase (280 Ma) (Conrad, 1974). During the Mesozoic, the geodynamic evolution of the region is marked by an Early Jurassic extensional phase (175–190 Ma) (Conrad, 1974) with correlative intrusion of dolerites. Radiometric dating of these rocks (Hailwood and Mitchell, 1971; Conrad, 1974) indicates a close relationship with Atlantic Ocean opening through the dislocations produced along a system of faults connected to the South Atlas fault zone (Sichler et al., 1980). The last tectonic event recognized in the region is the compressional phase of the Atlas (130–140 Ma) which produced a weak accentuation of folding and a change in tectonic direction in the northern part of the Béchar basin.

Methodology

Sampling sites were selected according to the occurrence of folded structures allowing the use of the fold test (McElhinny, 1964). The choice of

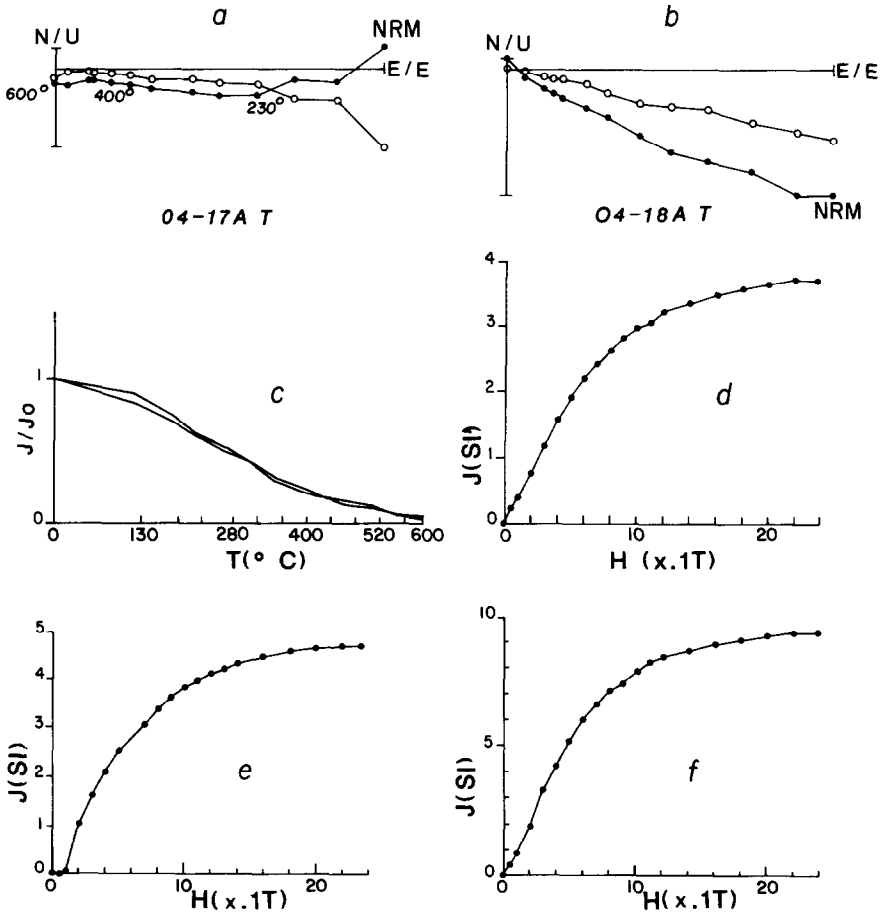


Fig. 2. Thermal demagnetization plots of specimens (a,b) from Béni-Abbès, showing two components of magnetization with an overlapping spectra of unblocking temperature or coercivity and their corresponding demagnetization curves (c) normalized to the initial NRM. Open circles: projection on vertical planes; dots: horizontal projection. IRM acquisition curves as a function of the inducing field from specimens of sites DA (d), CA (e) and CB (f).

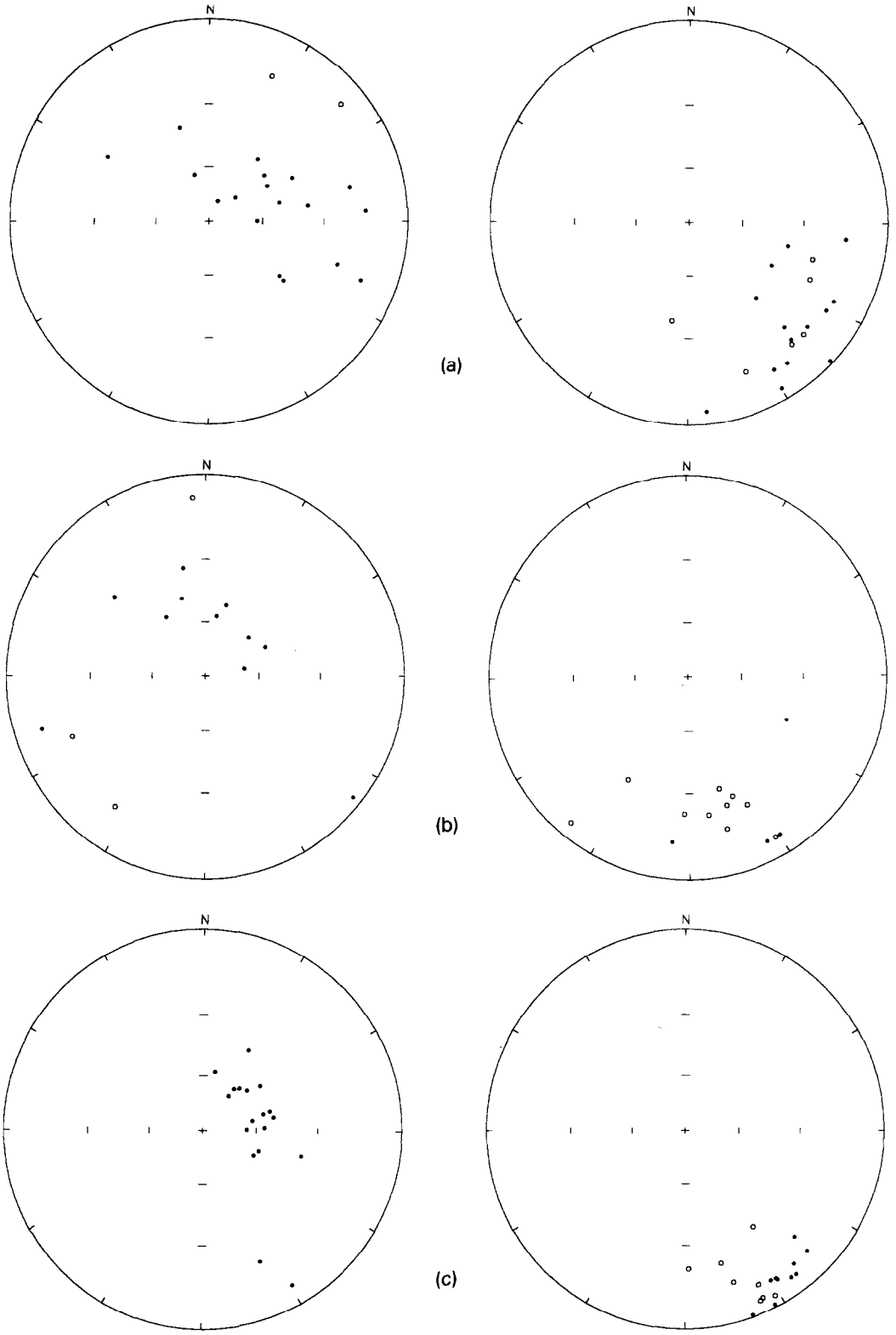


Fig. 3. Stereonet projections of the Lower Devonian limestones from Béni-Abbès specimens representing NRM (left) and characteristic (right) directions of magnetization before tectonic correction for sites CA (a), CB (b), DA (c). Solid symbols indicate lower hemisphere projection, open indicate upper hemisphere.

suitable formations was facilitated by the use of a field Spinner magnetometer. All the specimens were oriented with magnetic compass and often sun compass. Present declination is 4°30′W.

A CTF cryogenic magnetometer was used for magnetization measurements. The noise level of the instrument is about $5 \cdot 10^{-6}$ A/m. Thermal demagnetization was performed using a non-magnetic oven with a residual field in the cooling part of a few nannoteslas. Alternating field (A.F.) demagnetization was also used on some pilot samples. Direction and intensity of magnetization during demagnetization were analysed by vector diagram representation (Zijderveld, 1967). Isothermal Remanent Magnetization (IRM) acquisition in fields up to 2.4 T (IRM_{max}) was measured on at least two specimens from each site. Continuous thermal demagnetization of IRM_{max} was used to determine the different magnetic minerals by their Curie temperatures. This method was also used for the analysis of changes in the magnetic mineralogy which occur in some cases up to 520°C.

Analysis of paleomagnetic data

Devonian limestones

Three sites were sampled near Béni-Abbès in monoclinal and synclinal structures (Fig. 1a). These structures are formed by thick beds of limestones, moderately tilted (dip ≈ 10°), with an abundant fossil content (*Orthoceras* and *Goniatites*) indicating an Early Devonian age. IRM acquisition curves do not show saturation in field intensities of 2.4 T (Fig. 2d, e, f). The Natural Remanent Magnetization (NRM) intensities are in the order of 10^{-3} A/m, with a mean direction different from the present dipole field ($D = 70°$, $I = 50°$, $\alpha_{95} = 18°$) (Fig. 3). Typical vector diagrams (Fig. 2a, b) show in general two components of magnetization with close or overlapping spectra of unblocking temperature or coercivity. Characteristic directions isolated between 320°C and 660°C are summarized in Table 1. The fold test is inconclusive (Table 1). This might correspond to a post-folding remagnetization, probably carried by magnetite, but the fold test is not discriminatory because of the shallow dips of the structures. One could also content that a syntectonic acquisition had happened during the Permian. We have in fact noticed before that the folding did not occur before the Permian in this region. Nevertheless, the low inclination and the declination correspond to a Permian direction of remagnetization. Permian remagnetizations have been previously found (Hamzeh, 1973; Daly and Pozzi, 1976, 1977; Westphal, 1977; Martin et al., 1978; Westphal et al., 1979; Salmon et al., 1988a).

Griotte limestones

The Ben-Zireg anticline is located in the northern part of the Béchar basin (Fig. 1b). It occurs as an accentuated folding of Paleozoic strata between the Djebel Antar and the Djebel Béchar. The Paleozoic strata outcrop in a window of the Mesozoic cover. The core of the anticline is com-

TABLE 1

Mean directions from the Orthoceras-bearing limestones of Béni-Abbès sites *

Site	D_b (°)	I_b (°)	D_a (°)	I_a (°)	N	K	α_{95} (°)	Strike (°)	Dip (°)
CA	134.6	5.4	135.1	0	20	7	11.4	108	12
CB	167.9	−11.2	168.5	−0.8	14	7	14.1	288	12
DA	151.3	−1.9	151.1	4.7	18	25	6.6	290	10
Mean	149.3	−1.6	149.6	1.6	52	8	6.8		
Mean	151.2	−2.6	151.6	1.4	3	19(43)	18.4(16.9)		

* The directions (D, I) are marked before (b) and after (a) tectonic correction. The fold test is inconclusive (McElhinny, 1964). α_{95} and K are the usual Fisher's (1953) parameters. Numbers between parentheses are values after tilt correction. Tilt corrections (strike, dip) are also given for each site.

prised of Silurian and Lower Devonian deposits. This depressed area is dominated by thick limestone beds (100 m), light brown coloured, and dated as Famennian (Upper Devonian). 120 specimens have been cored in three sites (BZ, BE, BJ) representing significant dips (50° average) with various directions, and contouring the terminal anticline structure.

Rock magnetism

The three sites of griotte limestones show differences in NRM intensity. The mean NRM intensity is $0.4 \cdot 10^{-2}$ A/m, $0.25 \cdot 10^{-2}$ A/m and $0.5 \cdot 10^{-2}$ A/m respectively for the sites BZ, BJ and BE.

IRM acquisition curves are very similar for the three sites and an example is shown in Fig. 4. Saturation is not reached in an inducing field intensity of 2.4 T, leading to the conclusion that haematite is likely the dominant carrier of remanent magnetization. Thermal demagnetization of IRM obtained in 2.4 T given in Fig. 5 corroborates the presence of haematite and shows that other minerals are not detectable by their characteristic Curie temperature. The higher percentage of haematite with respect to Béni-Abbès is also underlined by the lack of A.F. demagnetization up to 0.025–0.035 T (Fig. 6).

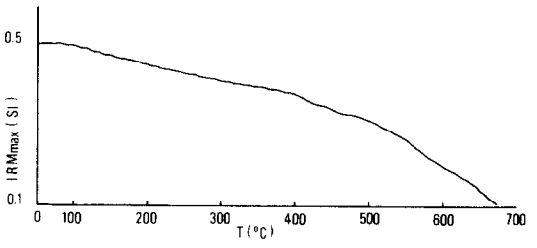


Fig. 5. Continuous thermal demagnetization curve of IRM_{max} of Ben-Zireg specimens indicating a Curie temperature at 680 °C corresponding to haematite.

Analysis of paleomagnetic directions

Site BZ. Stepwise thermal demagnetization curves are shown in Fig. 7 for site BZ. The directions isolated during thermal demagnetization show also different behaviour from one site to another (Fig. 8). Vector diagrams show successively the removal of a soft component close to the present dipole field at a temperature as low as 160°–200°C. Two characteristic components of magnetization using vector analysis (Kirschvink, 1980) can be isolated. The first one called intermediate unblocking temperature component (ITb) is isolated between 200° and 600°C. The second one, isolated up to 560°–600°C, is less well determined and is called high unblocking temperature component (HTb) (Fig. 7). The mean direc-

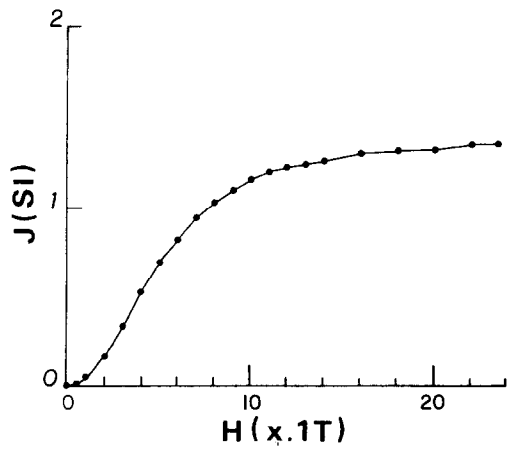


Fig. 4. IRM acquisition curve as a function of the inducing field indicating a haematite dominated carrier in the specimens of Ben-Zireg sites.

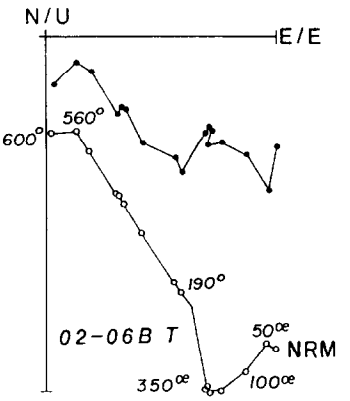


Fig. 6. Vector demagnetization for Ben-Zireg specimen (BE site) showing consecutive A.F. and thermal demagnetization with a high coercivity of grains up to 0.035 T (350 Oe) indicating the presence of haematite. Same notations as in Fig. 2.

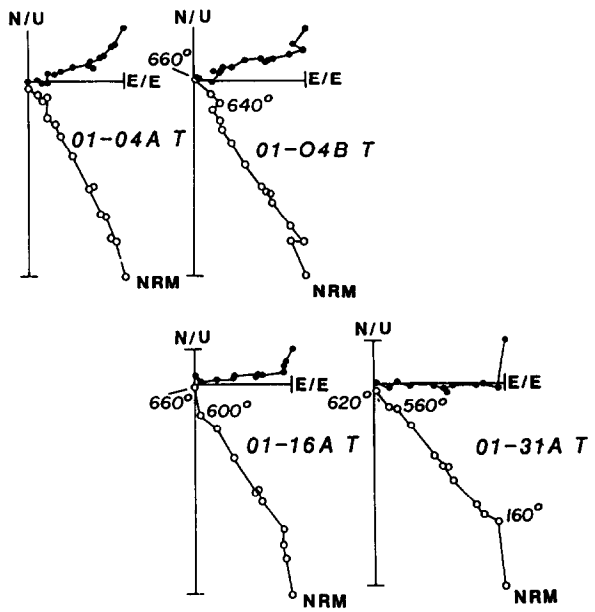


Fig. 7. Vector demagnetization diagrams (same notations as in Fig. 2) for Ben-Zireg specimens (BZ site), showing two components of magnetization: HTb and ITb components.

tions of magnetization for these two components are given in Tables 2 and 3.

Sites BJ and BE. Typical thermal demagnetization of specimens from sites BJ and BE are represented on vector demagnetization diagrams and on stereograms (see Fig. 10). Both representations show that the magnetization is the result of the superposition of two components with overlapping spectra of unblocking temperatures.

(1) *ITb component.* Some specimens do not show linear segments in vectorial projection dur-

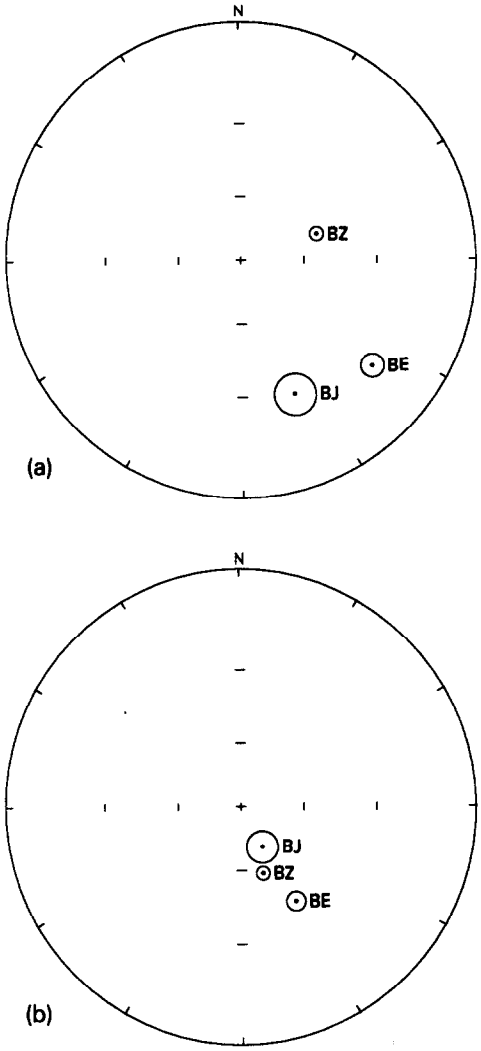


Fig. 8. Stereonet projections of the Famennian griotte limestones from Ben-Zireg Specimens (same conventions as in Fig. 3) representing the mean ITb components of magnetization for each site with their confidence level before (a) and after (b) tectonic correction. The fold test is positive (see Table 3).

TABLE 2

Mean HTb directions from the Beni-Zireg griotte limestones

Site	D_b (°)	I_b (°)	D_a (°)	I_a (°)	N	K	α_{95} (°)	Strike (°)	Dip (°)
BZ	90.8	51.7	159.4	45.6	47	92	2.1	120	53(50)
BE	132.7	35.7	169.8	49.8	26	22	5.9	170	40(27)
BJ	158.4	31.5	150.2	72.2	20	26	6.2	253	41(16)
Mean	120.6	46.6	161.5	52.5	93	10(23)	4.5(3.1)		
Mean	131.4	42.8	161.7	56.0	3	9(29)	26.6(15.1)		
Unfold			156.4	47.6	3	(8586)	(0.9)		
BZ + BE	107.9	48.0	162.8	47.1	73	16(40)	4.1(2.6)		

* Same notations as in Table 1. Dips between parentheses refer to partial unfolding (unfold) to release the HTb component of magnetization.

ing stepwise demagnetization between 160 °C and 500 °C and were discarded. The ITb component can be isolated by a least-squares method (Kirschvink, 1980) on 22 specimens (Table 3). The fold test is positive both for the mean resultant direction of the three sites BZ, BE and BJ (Fig. 8) and for the mean direction resultant of all specimens. Following these results, we consider the ITb direction as the older component of magnetization acquired *before any folding*. The magnetic carrier could be magnetite because of the unblocking temperature close to the Curie point of this mineral. The pre-folding direction of magnetization is shown in Table 3.

(2) *HTb component*. This component, still using vectorial analysis, has been isolated up to 560–600 °C in 46 specimens (Table 2). Dealing with the fold test, the result is likely positive for all specimens but inconclusive for the three sites. As the site BJ is more affected by overprints, if only sites BZ and BE are taken into account the improvement is also important for α_{95} (4.1 to 2.6) and k (16 to 40) leading likely to a positive fold test ($K_2/K_1 = 2.5$). But complete rejection of site BJ is unsatisfactory.

Progressive unfolding shows that the best cluster is reached with nearly a complete unfolding of BZ and partial unfoldings of the other sites (Fig. 9c, d and Table 2). This component of magnetization is likely considered as acquired after a partial folding of the structure. This assumption is suggested by the polyphased tectonism of the structure. In this case the magnetic carrier is clearly haematite (Fig. 5).

(3) *Great circles of remagnetization*. The difficulties encountered during the extraction of the ITb component in several specimens is due to the fact that after the removal of a soft component, the vector demagnetization diagram of these specimens shows curved segments (Fig. 10). In addition the directions are aligned on the stereonet along a great circle as a consequence of the demagnetization of one component more rapidly than another during stepwise heating (Fig. 10). But it is not always the case since some other specimens have a different behaviour.

The ITb component is progressively removed while, at a critical temperature (around 480–520 °C), the HTb (until now supposedly "harder") is then removed, leaving the ITb component in stable direction with an overlapping unblocking temperature (Fig. 10c; specimen 03-12B shows six demagnetization steps closely lumped before 600 °C). These results were therefore analysed using remagnetization circles (Halls, 1976, 1978).

The distribution of the directions of magnetization shows a clear elongation (Fig. 10). The great circles of remagnetization have been calculated by a least-squares method for several samples from sites BJ and BE, using temperature ranges where the evolution of the direction of magnetization is clearly the result of the predominant demagnetization of one of two components (between 100 °C and 660 °C). A selection of 22 specimens as noted in Table 4 has been used for the study of mixed remagnetization circles (Bailey and Halls, 1984), leading to three mean planes for both sites. The site BJ gives one mean plane called BJ_i but the site

TABLE 3

Mean ITb directions from the Ben-Zireg sites *

Site	D_b (°)	I_b (°)	D_a (°)	I_a (°)	N	K	α_{95} (°)
BZ	71.5	52.6	161.9	57.2	32	67	3.0
BE	128.9	19.4	149.8	40.7	11	123	3.8
BJ	158.5	27.6	152.3	68.4	11	32	7.4
Mean	109.6	44.1	154.9	54.3	54	6(24)	7.6(3.7)
Mean	127.1	38.0	154.3	55.6	3	5(32)	36.8(14.3)

* Same notations as in Table 1. Results show a positive fold test.

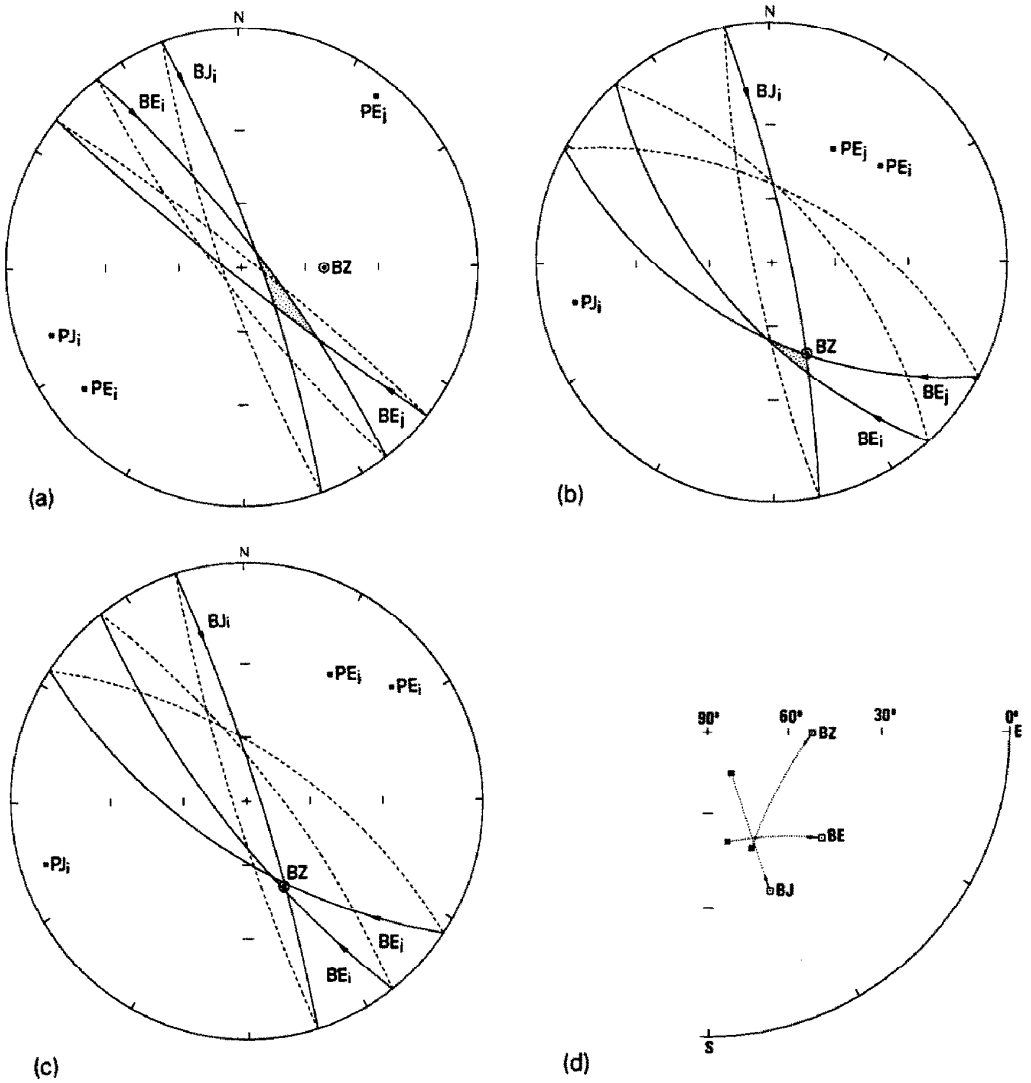


Fig. 9. Estimation of the confidence obtained by the intervals between extreme great circles of remagnetization corresponding to poles situated within the circles of confidence α_{95} of PE and PJ. a. Before tectonic correction: The circles of remagnetization are shown in full lines along with the shaded surface of intersection of the circles. b. After tectonic correction: The shaded surface is reduced and the mean direction of BZ moves to the intersection of the great circles. c. After partial unfolding: Notice the best convergence. d. Intersection of the small circles described by the mean direction of sites during folding (indicated by arrows). Open squares: sites before tectonic correction; close squares: after tectonic correction. The common intersection represents the best partial unfolding.

BE which is less homogeneous gives two separate mean planes called BE_i and BE_j. BE_i corresponds to stable inclination and increasing declination while for BE_j the declination is stable and the inclination increases during demagnetization.

The poles deduced from remagnetization circle of each specimen give a distribution allowing us to calculate a mean pole P which determines a mean circle and an angle of confidence α_{95}. For sites BE and NJ the mean poles PE_i, PE_j and PJ_i (Table 4) give reciprocally the mean circles of remagnetization for sites BE and BJ (Bailey and Halls, 1984).

Figure 9 gives the results of the three sites, three great circles of remagnetization for the sites BE and BJ compared to the mean characteristic direction of magnetization evidenced for the site

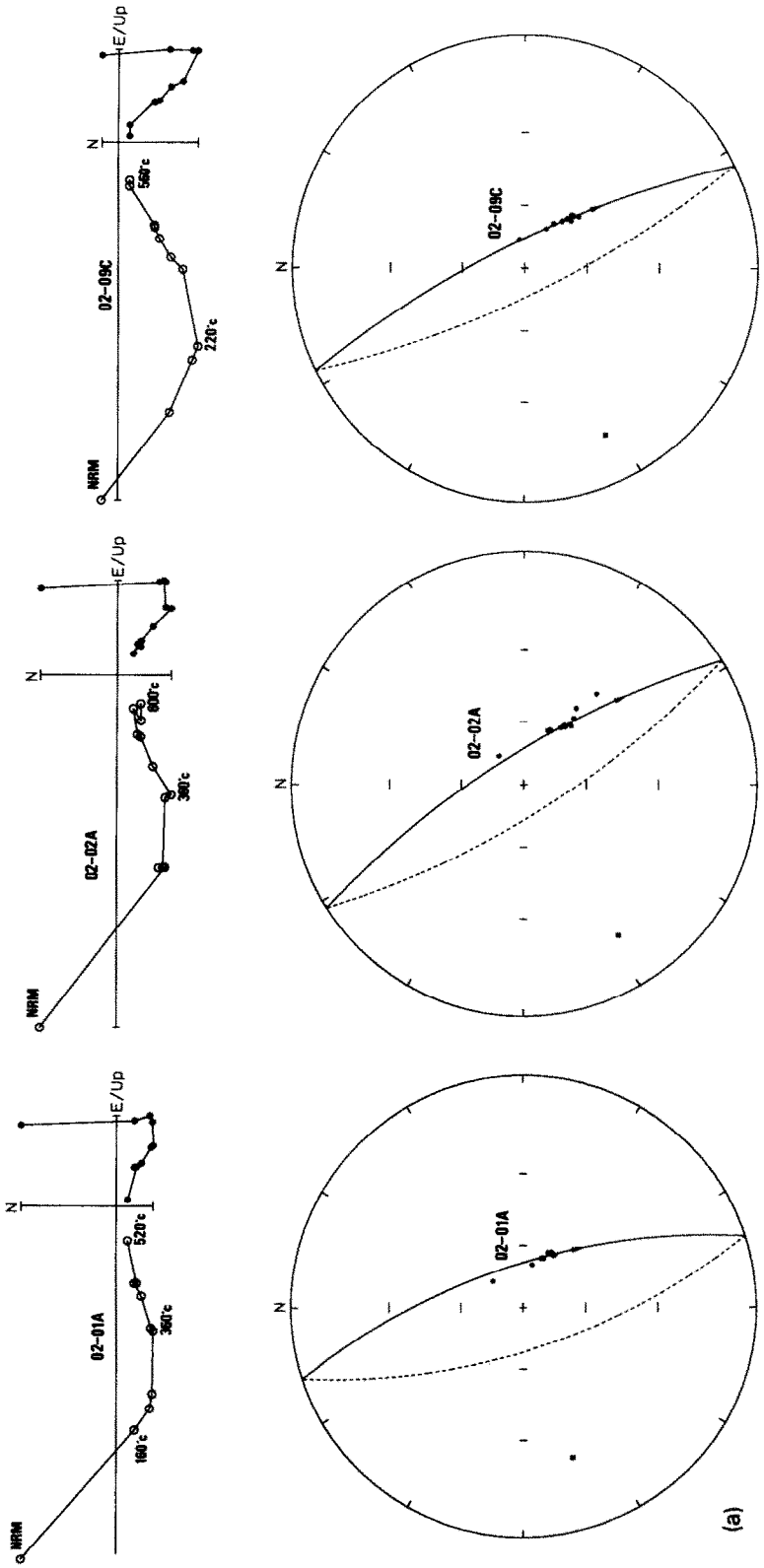


Fig. 10. Vector demagnetization diagrams (same notations as in Fig. 2) for BE (a, b) and BJ (c) specimens and their stereonet projection showing curved segments and repartition on great circles of remagnetization, before tectonic correction.

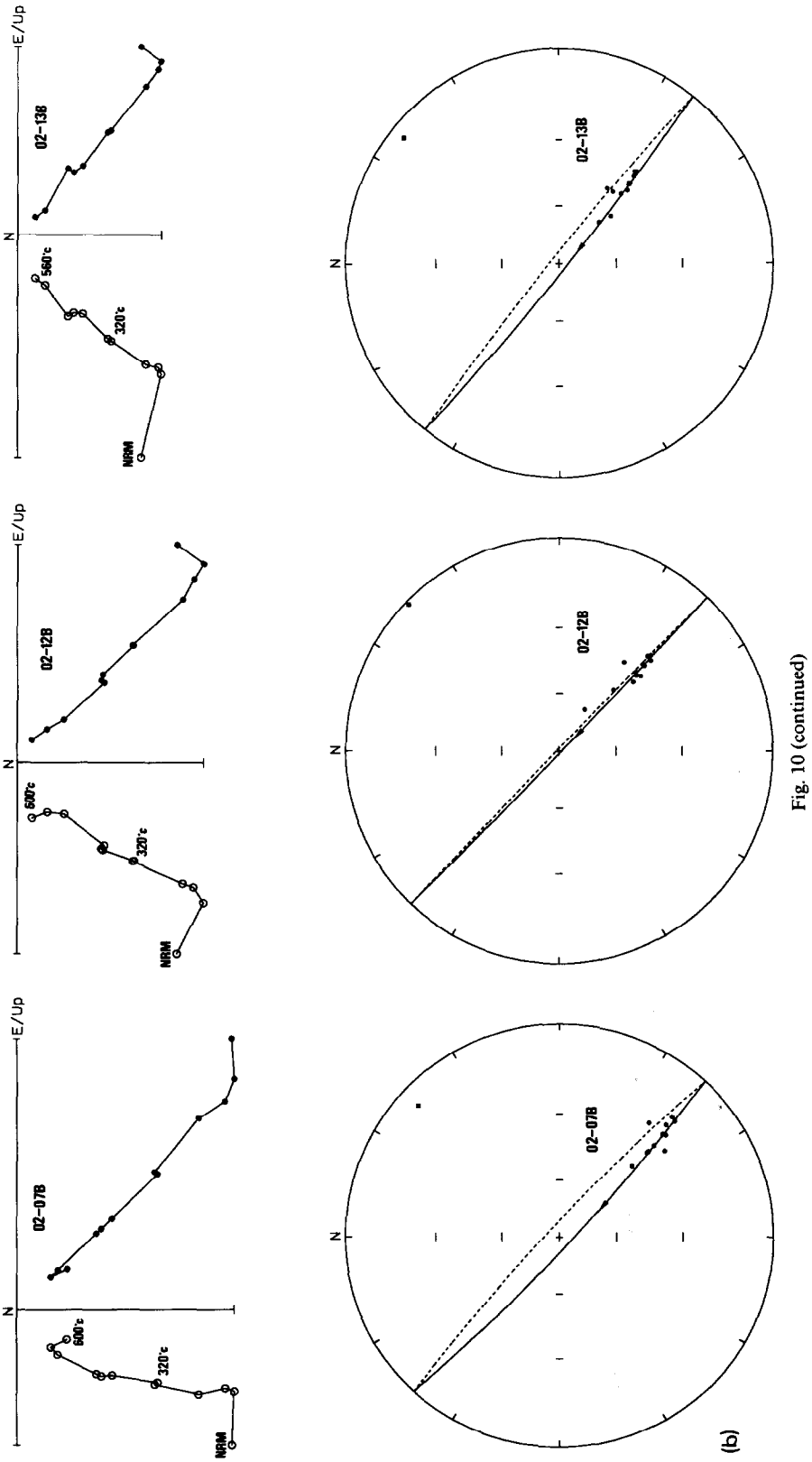


Fig. 10 (continued)

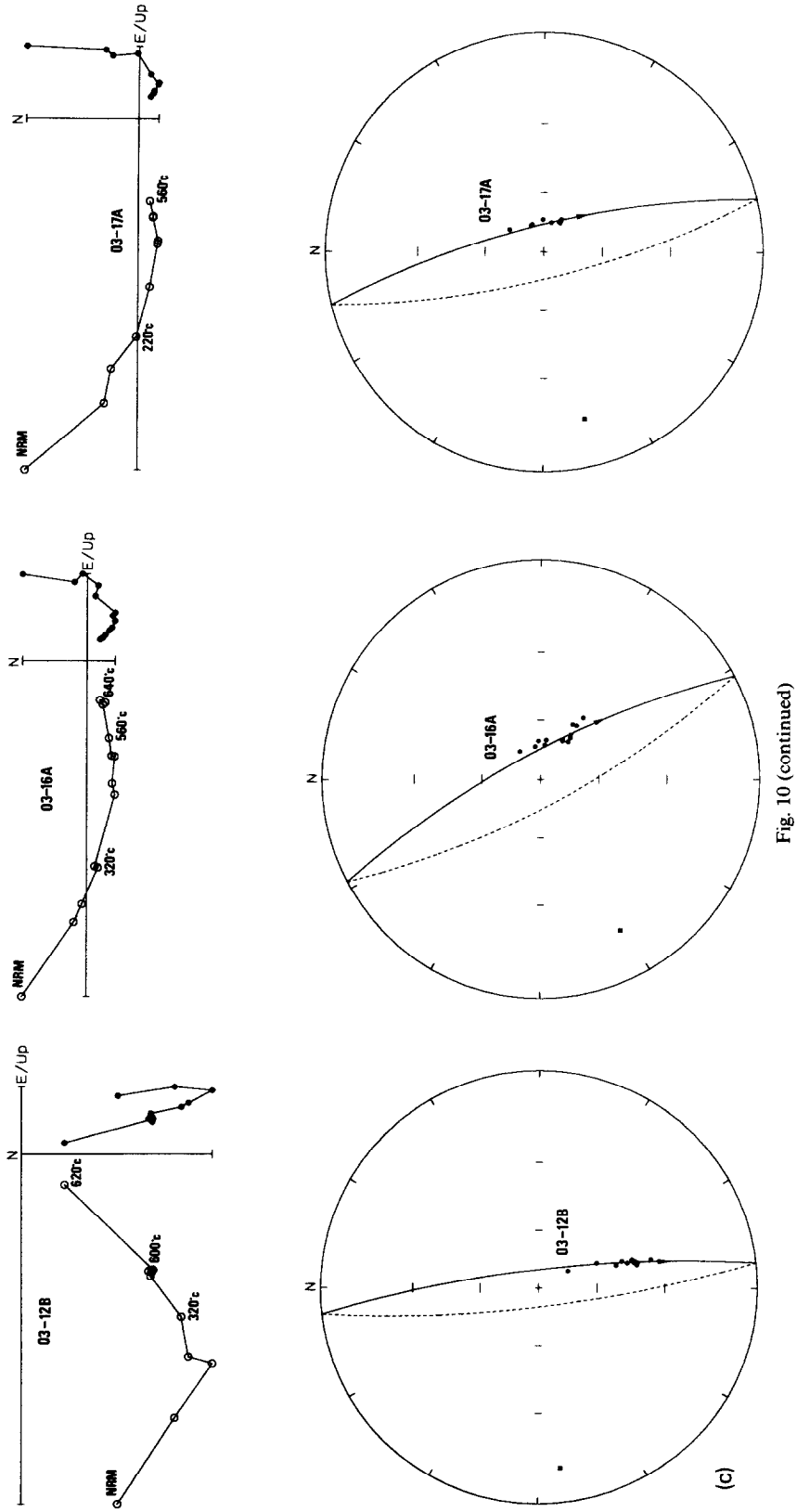


Fig. 10 (continued)

TABLE 4

Mean best fitted poles from the Ben-Zireg sites *

Site	D_b (°)	I_b (°)	D_a (°)	I_a (°)	N	K	α_{95} (°)
PJ_i	251	9	259	9	10	58	5.8
PE_i	233	10	49	27	7	85	5.7
PE_j	39	5	29	33	5	227	4.2

* Same notations as in Table 1. PE_i, PE_j and PJ_i are respectively the poles of the mean great circles BE_i, BE_j and BJ_i from sites BE and BJ.

BZ (HTb component). Figure 9a is before tectonic correction and Fig. 9b is after tectonic correction. The mean direction of BZ moves towards the intersection of the three great circles of remagnetization obtained for BE and BJ. These intersection points show a cluster with a total unfolding (Fig. 9b), but the convergence is better with an adequate partial unfolding (see Fig. 9c and Table 2) and gives the most accurate intersection of the great circles of remagnetization (Fig. 9c). The interpretation is that the anticline structure has been first partly folded at the beginning of the Hercynian collision and then remagnetized during the Early Carboniferous. This secondary direction of magnetization being carried by haematite is the so-called HTb component.

Time of acquisition of remanent magnetization

The ITb component of magnetization, i.e. the older direction given by analysis of the characteristic vectors, is a pre-folding acquisition of magnetization, therefore dated from the Famennian (Late Devonian).

The corresponding Late Devonian pole (365 Ma) for Africa is:

Paleolatitude = 34.8° S, VGP latitude = 19.2° S, VGP longitude = 19.8° E.

$dp = 3.7°$, $dm = 5.3°$

The convergence of remagnetization circles after partial unfolding corresponds to the secondary direction of magnetization. This HTb component of magnetization determined also by vectorial analysis could be dated from the Early Carboniferous (Tournaisian, 355 Ma) and the corresponding paleopole is:

Paleolatitude = 28.7° S, VGP latitude = 25.3° S, VGP longitude = 21.1° E.

$dp = 0.8°$, $dm = 1.2°$

Relation between folding and acquisition of remanence

Geological data indicate that Ben-Zireg area underwent a progressive deformation. Thus, the pre-folding direction of magnetization was likely acquired just before the first effects of deformation. The secondary direction of magnetization occurs when the structure is weakly folded with the following values:

Site	Strike	Dip
BZ	120°	3°
BE	170°	16°
BJ	253°	25°

Then the progressive continental shortening leads to an accentuation of the folding till the present values.

Discussion

The representation of paleolatitude–declination (Besse et al., 1984) consists of the calculation for the same area (here, Ben-Zireg) of the declination and the paleolatitude from known VGPs. Variations parallel to the paleolatitude axis correspond to latitudinal movements, those parallel to the declination axis give rotations (counter-clockwise

TABLE 5

Paleomagnetic data for the northwestern Sahara *

Site	Age (Ma)	N	D (°)	I (°)	α_{95} (°)	dp (°)	dm (°)	Lat. (°N)	Long. (°E)	Plat. (°N)	D (°)	Source
Béni-Abbès												
(29.7, −2.1)	270	3	149.6	1.6	6.8	3.5	6.9	−46.5	45.4	0.8	30.4	(a)
Hassi-Bachir												
(26.6, 1.8)	280	7	136.7	13.9	4.7	2.5	4.9	−36.2	59.3	−1	44.3	(b)
Abadla												
(31, −2.7)	300	13	129	11	6.0	3.1	6.1	−29.1	59.6	−5.5	51	(c)
Hassi Bachir												
(26.6, 1.8)	310	7	130.9	28	3.7	2.2	4.0	−26.8	56.7	−9	49.1	(b)
Ain Ech Chebbi												
(26.4, 0.4)	315	4	129.9	35.8	6.3	4.2	7.3	−22.9	52	−14.8	50.1	(b)
Ben-Zireg, HTb												
(31.9, −1.8)	355	3	156.5	47.6	0.9	0.8	1.2	−25.3	21.1	−28.7	23.6	(a)
Ben-Zireg, ITb												
(31.9, −1.8)	365	3	154.9	54.3	3.7	3.7	5.3	−19.2	19.8	−34.8	25.1	(a)

* Numbers between parentheses are locations of sites in degrees of latitude (°N) and longitude (°E) respectively. dp and dm are the semi-axes of the oval of 95% confidence about the mean pole. N, (D, I) represent respectively the number of sites, the characteristic directions of magnetization. Lat.(°N) and Long.(°E) are paleopoles. Plat.(°N) is the paleolatitude calculated and reduced at the site of Ben-Zireg. Ages are indicated in Ma (Haq and Van Eysinga, 1987). Declinations are specified with respect to the antipodal direction of the present dipole field. ITb and HTb are defined in the text. Sources: (a) this study; (b) Daly and Irving (1983); (c) Morel et al. (1981).

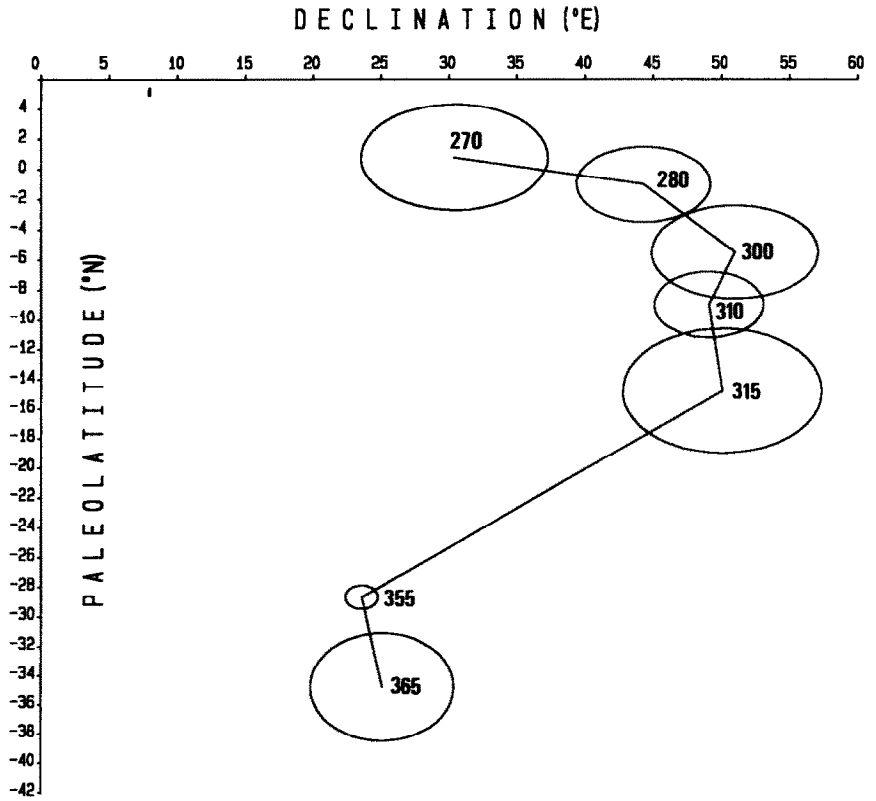
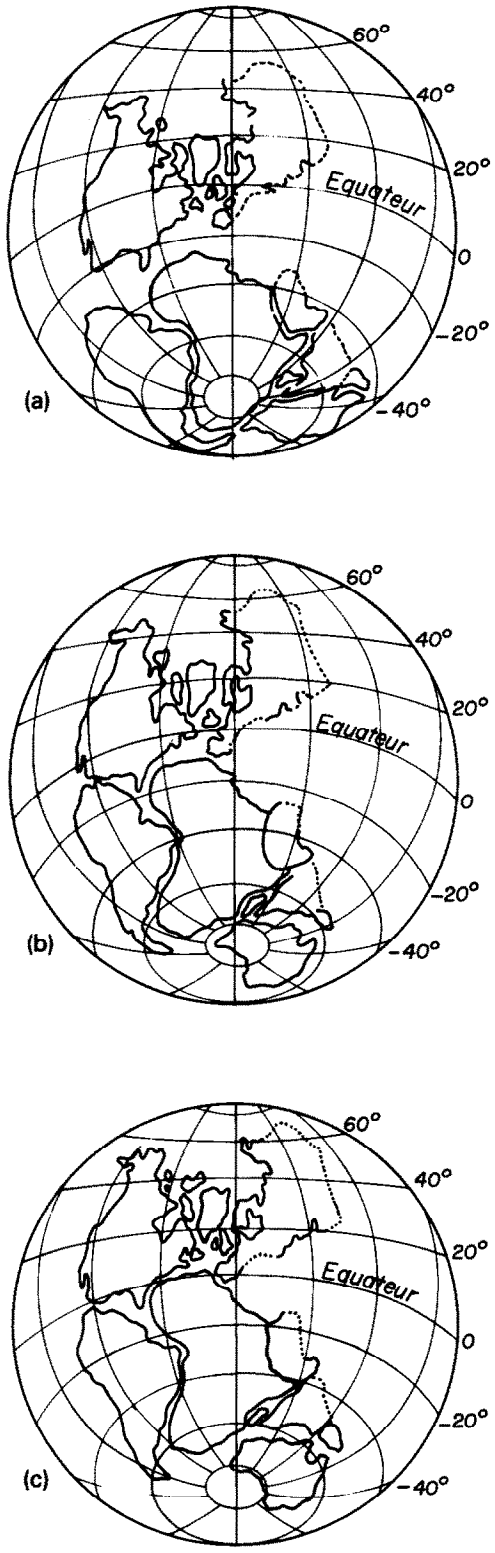


Fig. 11. Paleolatitude–declination diagram representative of the northwestern Sahara. Circles of confidence are drawn for the 95% probability. Numbers refer to the ages of the results (Table 5). Data are reduced to the Ben-Zireg locality.



to the west, clockwise to the east). The construction of the curve is based on the dating of the acquisition of the two directions of magnetization evidenced in the Ben-Zireg sites. We have computed the data relative to the northwestern Saharan region in Table 5. A curve (Fig. 11) deduced from these results gives the movements of Gondwana from Late Devonian to Permian times. The beginning of the collision probably occurred in the Late Devonian or just after, since flysch deposits uncomformably lie on Famennian griotte limestones (Pareyn, 1961). We note a S–N drifting of about 30° in latitude. At the beginning of the collision (i.e. between the Late Devonian and the Tournaisian) the movement of Gondwanaland seems to be faster (6 cm/yr) than at the end of the Middle Carboniferous (3.5 cm/yr). A clockwise rotation of about 25° is shown between the Tournaisian and the end of the Namurian (355–315 Ma), with a mean rate of 0.7°/Ma. Migration of Africa to the lower latitudes has also been established through sedimentological arguments (Sempère and Beaudoin, 1984) at Arlit (Niger), where Lower Carboniferous sedimentation is found to have occurred in a cold climate (Early Visean), in contrast to the warm and dry Late Carboniferous. Crustal shortening is computed from paleolatitudes of the northwestern Sahara supposing that Laurasia has encountered only a slight northward drift during Carboniferous times (McElhinny and Brown, 1973; Khramov et al., 1981; Ziegler, 1984). The mean value of 2500 km is determined from the dating of the remagnetization at Béni-Abbès between 270 and 290 Ma. This gives spreading rates comparable to present values. To show the movements indicated by this curve, we have represented three schemes (Fig. 12) corresponding to the Late Devonian, the Namurian and the Permian. The VGPs and the paleolatitudes were used to constrain the position of the continents. To restore the drifting of these blocks during the Paleozoic,

Fig. 12. Schemes representing the configurations of Gondwanaland and Laurasia corresponding to the Late Devonian (a), the Namurian (b) and the Permian (c). Note that the (c) configuration is equivalent to the A_2 Pangea model of Van der Voo and French (1974).

we used published paleomagnetic data. During the Late Devonian, tectonic activity had already begun on all sides of the Hercynian chain. One can think of "Andean" tectonism, but the collision between large blocks or between large blocks and microblocks seems to be more probable. Meanwhile, a separation still existed between Laurasia and Gondwana (Fig. 12a). Taking into account the shortening, we can easily estimate that the pre-Hercynian ocean was closing in this period (Fig. 12b). During the Carboniferous an important shortening of 2500 km occurs in close agreement with a kinematic model for the collision between Gondwanaland and Laurasia proposed by Lefort and Van der Voo (1981), while a simultaneous clockwise rotation of 25° with respect to the magnetic north for Africa is shown here. This rotation favours the hypothesis of a polyphased tectonism. The continents were sutured (Fig. 12c) according to the so-called Pangea A_2 model proposed by Van der Voo and French (1974). The crustal shortening continues but at a lower rate. A small counter-clockwise rotation of Africa must be taken into account.

Conclusion

The Late Devonian pole of this study, based on pre-folding remanent magnetization of well dated rocks seems to be the first to be known in this area. The other poles that we have recognized contribute to define a paleolatitude–declination curve for Gondwanaland which allows a better understanding of the evolution of the Hercynian collisional event. This curve shows a S–N shortening with a clockwise rotation of about 25° during a period of 40 Ma. The crustal shortening since the beginning of the collision is about 2500 km leading to a mean convergence rate of 3 cm/yr. At the Devonian–Carboniferous boundary our results seem to show that Gondwanaland was already much closer to Laurasia than shown in previous studies (Hailwood, 1974; Irving, 1977; Van der Voo, 1988) suggesting the closure of the pre-Hercynian ocean.

Permian remagnetization problems encountered show that the regions situated near the collisional front (internal zones) seem less affected by this phenomenon. The same conclusion arises from results obtained by Salmon et al. (1988b) westward from our study area in the Hercynian collisional zone where Carboniferous rocks near Erfoud (Morocco) have given primary directions of magnetization. On the other hand, the weakly deformed external zones are practically always remagnetized during the Kiaman (Early Permian) period, as shown by the example of the Béni-Abbès sites. Therefore, to find other Devonian and Carboniferous formations which are only weakly or not all remagnetized, it appears advisable to collect in the proximity of the Hercynian belt and at the Atlas flexure to augment the data for the Devonian and the Carboniferous of Africa.

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